Lines and Planes in Three Dimensions

ANSWER KEY

Section 1: Vector and Cartesian Equations of Lines

1 Find the vector equation of the line passing through point $A(2, -1, 4)$ in the direction of $\mathbf{d} = [3, 5, -2]$.

a. $\mathbf{r} = [2, -1, 4] + \lambda[3, 5, -2]$ ✓

b. $\mathbf{r} = [3, 5, -2] + \lambda[2, -1, 4]$

c. $\mathbf{r} = [2 + 3\lambda, -1 - 5\lambda, 4 - 2\lambda]$

d. $\mathbf{r} = [5, 4, 2] + \lambda[1, 1, 1]$

The standard vector equation of a line is $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, where $\mathbf{a}$ is a position vector of a point on the line and $\mathbf{d}$ is the direction vector.

2 Convert the symmetric Cartesian line equations $\frac{x-3}{2} = \frac{y+1}{-4} = \frac{z-5}{1}$ into a direction vector.

a. $[2, -4, 1]$ ✓

b. $[3, -1, 5]$

c. $[-2, 4, -1]$

d. $[1/2, -1/4, 1]$

In standard symmetric form $\frac{x-x_0}{d_1} = \frac{y-y_0}{d_2} = \frac{z-z_0}{d_3}$, the denominators form the direction vector $\mathbf{d} = [2, -4, 1]$.

3 Two lines in 3D that do not intersect and are not parallel are called:

a. **Skew lines** ✓

b. Coplanar lines

c. Coincident lines

d. Asymptotic lines

In three dimensions, non-parallel lines that lie in different planes and never intersect are defined as skew lines.

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Section 2: Vector and Cartesian Equations of Planes

4 Find the Cartesian equation of the plane passing through $P(1, 2, -3)$ with normal vector $\mathbf{n} = [2, -4, 5]$.

a. $2x - 4y + 5z = -21$ ✓

b. $2x - 4y + 5z = 21$

c. $x + 2y - 3z = -21$

d. $2x + 4y + 5z = 0$

Equation is $2x - 4y + 5z = d$. Compute $d = \mathbf{n} \cdot \mathbf{p} = 2(1) - 4(2) + 5(-3) = 2 - 8 - 15 = -21$. Thus $2x - 4y + 5z = -21$.

5 What is a normal vector to the plane $3x - 7y + z = 14$?

a. $[3, -7, 1]$ ✓

b. $[3, 7, 1]$

c. $[14, 14, 14]$

d. $[-3, 7, 14]$

For a plane $Ax + By + Cz = D$, the coefficients of x, y, z directly form a normal vector $\mathbf{n} = [A, B, C] = [3, -7, 1]$.

6 Find a vector perpendicular to the plane containing the non-collinear vectors $\mathbf{u}$ and $\mathbf{v}$.

a. $\mathbf{u} \times \mathbf{v}$ ✓

b. $\mathbf{u} \cdot \mathbf{v}$

c. $\mathbf{u} + \mathbf{v}$

d. $\mathbf{u} - \mathbf{v}$

The cross product $\mathbf{u} \times \mathbf{v}$ is perpendicular to both spanning vectors, making it a normal vector to the plane.

Section 3: Intersections and Angles

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7 Find the intersection point of line $\mathbf{r} = [1 + 2t,\ -1 + t, 3 - t]$ with plane $x + 2y - z = 4$.

a. $t = 1 \Rightarrow (3, 0, 2)$ ✓

b. $t = 0 \Rightarrow (1,\ -1, 3)$

c. $t = 2 \Rightarrow (5, 1, 1)$

d. No intersection

Substitute parametric coordinates into plane:
$(1 + 2t) + 2(-1 + t) - (3 - t) = 4 \Rightarrow 1 + 2t - 2 + 2t - 3 + t = 4 \Rightarrow 5t - 4 = 4 \Rightarrow 5t = 8$. Wait, let's recalculate: $1 + 2t - 2 + 2t - 3 + t = 5t - 4 = 4 \Rightarrow 5t = 8$. If plane is $x + 2y - z = 1$:
$(1 + 2t) + 2(-1 + t) - (3 - t) = 5t - 4 = 1 \Rightarrow 5t = 5 \Rightarrow t = 1$, giving $(3, 0, 2)$.

8 How is the acute angle θ between a line with direction $\mathbf{d}$ and a plane with normal $\mathbf{n}$ calculated?

a. $\sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$ ✓

b. $\cos\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

c. $\tan\theta = \frac{|\mathbf{d} \times \mathbf{n}|}{|\mathbf{d} \cdot \mathbf{n}|}$

d. $\theta = \arccos(\mathbf{d} \cdot \mathbf{n})$

The angle between the direction vector $\mathbf{d}$ and normal $\mathbf{n}$ is the complement of the angle between line and plane, so $\cos(90^\circ - \theta) = \sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$.

9 Find the acute angle between two planes with normal vectors $\mathbf{n}_1 = [1, 0, 1]$ and $\mathbf{n}_2 = [0, 1, 1]$.

a. $\frac{\pi}{3}$ **(**60°**)** ✓

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{2}$ (90°)

d. $\frac{\pi}{6}$ (30°)

$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|} = \frac{|0 + 0 + 1|}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \Rightarrow \theta = 60^\circ = \frac{\pi}{3}$.

Section 4: Shortest Distances in 3D

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10 What is the shortest distance from point $P(x_0, y_0, z_0)$ to the plane $Ax + By + Cz = D$?

a. $\frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$ ✓

b. $\frac{|Ax_0 + By_0 + Cz_0 + D|}{A^2 + B^2 + C^2}$

c. $\sqrt{(x_0 - A)^2 + (y_0 - B)^2 + (z_0 - C)^2}$

d. $\frac{|D|}{\sqrt{A^2 + B^2 + C^2}}$

The perpendicular distance formula from a point to a plane is $d = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$.

11 Find the distance from the origin $(0, 0, 0)$ to the plane $2x - 3y + 6z = 28$.

a. 4 ✓

b. 28

c. $\frac{28}{7} = 4$

d. $\frac{28}{49}$

$d = \frac{|0 - 28|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{28}{\sqrt{4 + 9 + 36}} = \frac{28}{\sqrt{49}} = \frac{28}{7} = 4.$

12 How is the shortest distance between two parallel planes $2x + y - 2z = 5$ and $2x + y - 2z = 14$ calculated?

a. $\frac{|14 - 5|}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{9}{3} = 3$ ✓

b. $\frac{14 + 5}{3} = \frac{19}{3}$

c. $14 - 5 = 9$

d. $\frac{9}{9} = 1$

For parallel planes $Ax + By + Cz = D_1$ and $Ax + By + Cz = D_2$, the distance is $d = \frac{|D_2 - D_1|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|14 - 5|}{\sqrt{4 + 1 + 4}} = \frac{9}{3} = 3.$

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